

Electron-molecule scattering via R -matrix variational algorithms on a quantum computer

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Electron-molecule collisions play a central role in both natural processes and modern technological applications, particularly in plasma processing. Conventional computational strategies such as the R -matrix method have been widely adopted yet encounter significant scaling challenges in treating more complex systems. In this work we present a quantum computational approach that utilizes the variational quantum eigensolver (VQE) and variations thereof to overcome these limitations. We explore a number of methods, including the use of number projection operators and simultaneous optimization. We demonstrate the feasibility of our method on a model problem of electron scattering from the hydrogen molecule, with numerical results obtained using a noiseless classical simulator. We recover the full spectrum of the Hamiltonian within a chosen symmetry sector. Moreover, the optimal circuit parameters directly encode the R -matrix boundary amplitudes needed for subsequent scattering computations. We formulate the R -matrix inner-region problem for electron-molecule scattering in a variational quantum-computing framework.

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I. INTRODUCTION

Electron-molecule collisions drive a myriad of natural processes and are instrumental in modern technologies, especially in plasma processing [1–3]. Studies of these processes increasingly rely on quantum mechanical calculations rather than on experiments [4]. However, for many problems of interest the computational resources needed for highly accurate solutions scales so drastically that approximate methods are routinely employed.

A host of quantum algorithms have been developed over the years. The idea of simulating quantum systems on a computer originated with Feynman [5] and was later formalized in Lloyd's seminal work on universal quantum simulators [6]. Early proposals for quantum phase estimation applied to chemistry were developed by Abrams and Lloyd [7,8] and

led to the first full simulation of molecular energies on a quantum computer by Aspuru-Guzik *et al.* [9]. More recently, in the noisy intermediate-scale quantum (NISQ) era, the variational quantum eigensolver (VQE) [10]—later refined by McClean *et al.* [11]—has become a popular method for electronic structure calculations. Early experimental demonstrations include implementations using photonic platforms [12] and superconducting qubits [13]; further experiments on trapped ions have also been reported [14]. Reviews covering both methodology and applications now abound in the literature [15–19]. Despite their promise, variational quantum algorithms (VQAs) can face significant optimization challenges. More recent hardware demonstrations have scaled and improved VQE-based quantum chemistry implementations, including optimized unitary coupled-cluster approaches [20] and trapped-ion demonstrations targeting relativistic molecular properties [21]. In particular, the training landscape may exhibit barren plateaus, where gradients concentrate and become exponentially small with system size for broad classes of *Ansätze* and cost functions, severely hindering trainability [22]. Relatedly, recent work has highlighted that circuit structures which provably avoid barren plateaus can, in some settings, also imply restrictions that enable efficient classical simulation of the loss (or parts of it), sharpen-

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ing the need to understand when trainability translates into genuine quantum advantage [23]. These considerations motivate the use of problem-structured, symmetry-preserving, and relatively shallow circuit constructions, and cost functions that remain close to physically meaningful objectives (here, sequential energy minimization within a fixed symmetry sector).

In parallel, quantum algorithms for scattering have been investigated. Jordan *et al.* proposed quantum algorithms for calculating scattering amplitudes in quantum field theories [24–26], while recent proposals [27–30] demonstrate the progress toward simulating scattering on quantum devices.

While quantum algorithms for scattering have been explored in a variety of contexts, prior work has not addressed electron-molecule collisions. For example, Xing *et al.* used a variational quantum linear solver within the Kohn variational framework to compute multichannel S matrices for atom-molecule collisions [31], and grid-based first-quantized simulations have treated electron scattering and ionization primarily in atomic model systems [32]. Other demonstrations on quantum hardware have extracted two-body scattering phase shifts in effective field-theory or gauge-theory settings [30,33]. By contrast, here we formulate and demonstrate a quantum algorithm for the electron-molecule case, mapping the R -matrix inner-region problem to qubits and variational circuits.

We enforce a fixed spin symmetry through a two-qubit Givens block in a variational scattering *Ansatz* and read out scattering boundary amplitudes directly from the optimized angles, a design not used in prior QC scattering demonstrations [31,33].

Simultaneous optimization of multiple eigenstates has been explored in the subspace-search variational quantum eigensolver (SSVQE) [34] and multistate contracted variational quantum eigensolver (MCVQE) [35]. The present construction differs from these subspace methods in three respects relevant to the R -matrix problem. (i) We use a fixed-order $SO(k)$ Givens cascade and optimize it sequentially: once the parameters fixing one low-energy state have been determined, the remaining free parameters act only within the residual orthogonal subspace, so the next state is isolated by a new minimization of the expectation value of H . This staged block diagonalization differs from the single weighted-subspace optimization used in the all-states SSVQE construction and from the contracted-subspace extraction used in MCVQE. (ii) The SSO objective requires only first-moment Hamiltonian expectation values. In contrast to the variance and folded-Hamiltonian alternatives considered below, it avoids H^2 -type observables and combines naturally with the coherent-summation readout on the selector register to reduce the number of distinct expectation values required. (iii) The optimized $SO(k)$ network acts directly on the channel and L^2 trial space of the R -matrix expansion, so its open-channel amplitudes give the coefficients a_{ijk} from which the boundary amplitudes $w_{ik(a)}$ are obtained. Thus the same variational parameters that determine the inner-region eigenstates also provide the scattering-specific quantities needed for the outer-region calculation.

This structure lets us avoid squared (H^2) terms and pair with a coherent-summation readout over the selector reg-

ister to cut the number of distinct expectation values. The optimized $SO(k)$ network determines the open-channel coefficients that enter the R -matrix boundary amplitudes, giving problem-native outputs beyond generic multistate VQE.

In this work we:

(1) Present a quantum-algorithm framework for the R -matrix inner-region problem, mapping the $(N + 1)$ -electron Hamiltonian to qubits via a Jordan-Wigner transformation and explicitly enforcing occupancy constraints for continuum orbitals.

(2) Employ the qubit representation of the number-projection operator in the variational circuit—based on the construction in Ref. [36]—to ensure that at most one electron occupies any continuum orbital.

(3) Develop a SSO protocol that decomposes the required $SO(k)$ rotations into a fixed cascade of two-qubit Givens rotations (i.e., rotations that mix the $|01\rangle$ and $|10\rangle$ subspace), block-diagonalizing H one eigenstate at a time and enforcing orthogonality by construction, without the need to introduce penalty terms, overlap measurements, or subspace diagonalization.

(4) Introduce a coherent-summation measurement scheme for the target-state selector register, which combines ancilla-based Hadamard gates with system-Hamiltonian measurements to reduce the total number of distinct expectation values required.

(5) Demonstrate our approach on a model H_2 scattering calculation, validating the algorithm and showing that the optimal circuit parameters determine the R -matrix boundary amplitudes needed for subsequent scattering analysis. The numerical results reported were obtained on a noiseless classical simulator (i.e., no device noise is included). Unless stated otherwise, expectation values are evaluated exactly (state-vector expectation values) rather than by finite-shot sampling, so there is no statistical measurement uncertainty to report as error bars.

II. THE R -MATRIX METHOD AND MOLECULAR HAMILTONIAN

The R -matrix method was originally developed to study nuclear [37] and then atomic and molecular collisions [37,38]. It is now [39] in widespread use to study electron collisions for a whole range of molecular problems and is also employed to study photoionization and molecules in strong fields [40]. The formulation of the R -matrix method is to divide the problem into a scattering-energy-independent inner region which, for electron-molecule collisions, has strong similarities with problems of quantum chemistry [39] and an energy-dependent outer region. Complicated outer-region problems can be computationally expensive but are well suited for modern, massively parallel computations [41]. Conversely, the inner region, which is generally treated using variational methods, shows scaling similar to that encountered in the quantum chemistry electronic structure problem but with extra complications which are discussed below.

In this paper, we propose and provide formulations for solving the R -matrix inner-region problem using an extension of the VQE framework. Two of us developed an R -matrix with pseudostates (RMPS) procedure for studying electron-

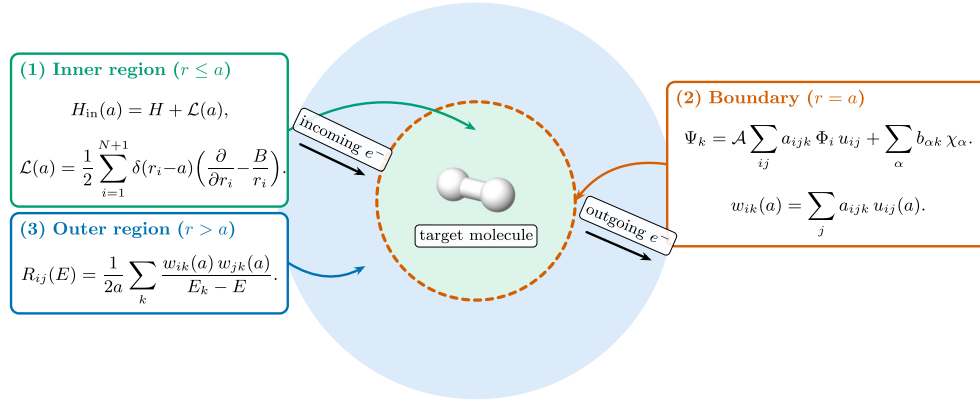


FIG. 1. Schematic of the *R*-matrix partition for electron-molecule scattering. The inner region is the sphere of radius a containing the molecular target, where the Hermitian inner-region operator $H_{\text{in}}(a) = H + \mathcal{L}(a)$ is solved. At the boundary $r = a$, the inner-region eigenstates define the surface amplitudes $w_{ik}(a)$ entering the *R*-matrix $R_{ij}(E)$, which provides the matching condition for propagation in the outer region.

molecule scattering [42,43] which has shown to provide robust numerical solutions to challenging scattering problems [44,45]. However, the computational demands of this RMPS procedure scale so sharply with the number of electrons that need to be considered that it remains largely unused despite ever-increasing demands for accurate cross sections. A quantum computational implementation of the *R*-matrix method therefore has very significant potential uses and benefits.

The *R*-matrix formalism addresses scattering problems by partitioning space into an inner region and an outer region [39] (Fig. 1). In the inner region, the full many-electron Hamiltonian is solved within a confined volume that encompasses the molecular target, allowing for a detailed treatment of electron correlation, exchange, and short-range interactions.

In our work we use the standard second-quantized representation of the molecular Hamiltonian, which is given by

$$H = \sum_{pq} h_{pq} a_p^\dagger a_q + \frac{1}{2} \sum_{pqrs} h_{pqrs} a_p^\dagger a_q^\dagger a_r a_s + h_{\text{nuc}}, \quad (1)$$

where h_{pq} are the one-electron integrals (accounting for kinetic energy and electron-nucleus attraction), h_{pqrs} are the two-electron integrals (accounting for electron-electron repulsion), and h_{nuc} represents the nuclear repulsion energy. In the inner region these integrals are computed over a finite volume (see below).

In conventional molecular electronic structure calculations the one- and two-electron integrals are computed over all space using basis functions that are typically Gaussians or Slater-type orbitals. For example, the one-electron integrals have the general form

$$h_{pq} = \int \phi_p^*(\mathbf{r}) h(\mathbf{r}, -i\hbar\nabla) \phi_q(\mathbf{r}) d\mathbf{r}, \quad (2)$$

and the two-electron integrals are given by

$$h_{pqrs} = \iint \phi_p^*(\mathbf{r}_1) \phi_q^*(\mathbf{r}_2) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \phi_r(\mathbf{r}_1) \phi_s(\mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2. \quad (3)$$

These integrals account for the full spatial extent of the electronic wave functions, ensuring that both short- and long-range interactions are included.

In the *R*-matrix inner-region problem one uses the same physical operators (kinetic energy, electron-nucleus attraction, and electron-electron repulsion) but solves them on a finite domain: all coordinate integrals are taken only over the inner region, typically a sphere of radius a that encloses the target. In this sense, the differential and Coulomb operators are unchanged; what changes is the region of integration and the choice of basis functions and boundary treatment at $r = a$.

Accordingly, the one-electron integrals are evaluated as

$$h_{pq} = \int_{r \leq a} \phi_p^*(\mathbf{r}) h(\mathbf{r}, -i\hbar\nabla) \phi_q(\mathbf{r}) d\mathbf{r}, \quad (4)$$

with basis functions adapted to the finite domain. In particular, continuum orbitals are constructed for the inner region and are allowed to have nonzero amplitude at the boundary; the boundary condition at $r = a$ is commonly enforced by augmenting the inner-region Hamiltonian with a surface (Bloch) operator, which restores Hermiticity on the truncated domain and provides the correct matching conditions [46]. Long-range interactions beyond $r = a$ are then handled separately in the outer region.

Each $(N+1)$ -electron eigenfunction Ψ_k of the resulting Hamiltonian (associated with an energy E_k) can be expanded as

$$\Psi_k = \mathcal{A} \sum_{ij} a_{ijk} \Phi_i u_{ij} + \sum_{\alpha} b_{\alpha k} \chi_{\alpha}, \quad (5)$$

where $\mathcal{A}$ is the antisymmetrization operator which enforces antisymmetry of the $(N+1)$ -electron wave function under the exchange of any two electrons, $\Phi_i(x_1, \dots, x_N)$ are the N -electron target eigenstates (indexed by i), $u_{ij}(x_{N+1})$ are continuum orbitals for channel (i, j) , a_{ijk} are the open-channel coefficients (with indices corresponding to target i , continuum j , and state k), and χ_{α} are purely bound (L^2) configurations, with L^2 coefficients $b_{\alpha k}$.

On the boundary $r = a$, only the continuum part contributes, defining the surface amplitudes. The *R*-matrix on the boundary is then given by the expansion

$$R_{ij} = \frac{1}{2a} \sum_k \frac{w_{ik}(a) w_{jk}(a)}{E_k - E}, \quad (6)$$

where the boundary amplitude $w_{ik}(a)$ is defined as

$$w_{ik}(a) = \sum_j a_{ijk} u_{ij}(a), \quad (7)$$

and the sum is taken over a given channel.

These equations encapsulate the influence of the short-range (inner) region on the scattering process and introduce the energy dependence of this process. In the outer region—where the wave function is determined by a simpler long-range potential—the R -matrix serves as the matching condition; in practice, outer-region solutions are usually obtained by propagating this R -matrix [39].

A possible workflow is as follows:

(1) A classical code (e.g., UKRmol+) computes the inner-region integrals, continuum orbitals, and boundary values $u_{ij}(a)$; a preceding quantum target-state routine provides the retained target eigenfunctions $\{\Phi_i\}$.

(2) The quantum device runs the variational inner-region algorithm for the $(N+1)$ -electron problem and returns optimized angles $\{\theta_{ij}^*\}$, thereby fixing the mixing matrix $U(\theta^*)$ and the channel coefficients a_{ijk} .

(3) A classical postprocessing script evaluates the required matrix elements of $U(\theta^*)$ (equivalently, translates θ^* into a_{ijk}), then combines a_{ijk} with the known boundary values $u_{ij}(a)$ to obtain the surface amplitudes $w_{ik}(a)$ and hence the R -matrix.

(4) A classical outer-region solver propagates the R -matrix [or equivalently uses the $w_{ik}(a)$] to obtain scattering observables such as cross sections.

A single solution obtained in the inner region gives rise to solutions for a broad range of scattering energies. This occurs because the inner-region eigenfunctions, which incorporate all the complex short-range correlation effects, are used to define boundary conditions that must be met by many different asymptotic states. Furthermore, the process of solving the inner-region problem can be computationally very demanding, as it involves the diagonalization of matrices whose size increases rapidly with the level of detail needed to accurately describe electron correlation.

There are a number of key differences between the R -matrix Hamiltonian described above and the standard molecular Hamiltonian solved by standard quantum chemistry problems. As already mentioned, it only considers a finite region, and an important assumption of the R -matrix method is that the target wave functions have zero amplitude outside this region. Conversely, the continuum functions do have amplitude on the boundary but cannot in total contain more than a single electron. In quantum-chemistry VQE one typically targets only the ground state; by contrast, R -matrix scattering requires many eigenstates in the given symmetry sector, including numerous excited pseudostates or continuum-type states [47]. This makes a naïve one-eigenstate-at-a-time approach costly. The method developed below is designed to accommodate these changes.

III. METHOD

We map the second-quantized Hamiltonian for the inner-region problem to a qubit Hamiltonian through the standard Jordan-Wigner transformation to obtain a qubit Hamiltonian

suitable for quantum computation [48,49], although in further work we plan to employ the more compact encodings we have developed [50].

Here and throughout Sec. III, H denotes the inner-region Hermitian operator $H_{\text{in}}(a) = H + \mathcal{L}(a)$ (Fig. 1), i.e., the Hamiltonian built from finite-domain integrals together with the Bloch surface operator that restores Hermiticity on $r \leq a$.

As with other VQA-based approaches, the practical performance of our method depends on optimization landscape properties and on the susceptibility to vanishing gradients. Barren plateaus can arise for certain *Ansätze* or cost choices and can make training intractable as problem size grows [22]. Moreover, the relationship between provable trainability guarantees and classical simulability is subtle, and understanding it is important when assessing where quantum resources provide an advantage [23]. In this work we mitigate these issues pragmatically by using a structured Givens-rotation network with explicit symmetry constraints and by employing sequential subspace optimization with a simple $\langle H \rangle$ objective, though a systematic scaling study of trainability for larger active spaces remains for future work.

A. Number projection operator

In the inner-region formulation the Hamiltonian must be projected onto a subspace in which at most one electron occupies the continuum orbitals. In practice, we define a projection operator P such that

$$P = P_{t,N+1} P_{c,0} + P_{t,N} P_{c,1}, \quad (8)$$

where $P_{t,n}$ projects onto configurations with n electrons in the target orbitals and $P_{c,m}$ projects onto configurations with m electrons in the continuum. Thus, $P_{t,N+1} P_{c,0}$ enforces all $N+1$ electrons bound and none in the continuum, while $P_{t,N} P_{c,1}$ enforces one electron in the continuum and N bound.

In the Jordan-Wigner mapping, $P_{t,N}$ can be expressed as a polynomial of Pauli-Z operators on the target-orbital qubits that enforces Hamming weight N . Detailed expressions for the qubit representation of the number projection operators have been provided by Picozzi [36].

Our *Ansatz* enforces the single-continuum-occupancy constraint structurally, so the prepared state obeys

$$|\psi(\theta)\rangle = P|\psi(\theta)\rangle, \quad (9)$$

and plain energies can be evaluated as

$$\langle \psi(\theta) | H | \psi(\theta) \rangle = \langle \psi(\theta) | PHP | \psi(\theta) \rangle. \quad (10)$$

However, for second-moment-based costs (i.e., variance, folded Hamiltonian) one needs the expectation of the square of the projected Hamiltonian $\langle \psi(\theta) | (PHP)^2 | \psi(\theta) \rangle$.

On any state with $|\psi(\theta)\rangle = P|\psi(\theta)\rangle$ this second moment can be evaluated as

$$\langle \psi(\theta) | (PHP)^2 | \psi(\theta) \rangle = \langle \psi(\theta) | PHPHP | \psi(\theta) \rangle \quad (11)$$

$$= \langle \psi(\theta) | HPH | \psi(\theta) \rangle, \quad (12)$$

where, in general, $\langle \psi | PH^2 P | \psi \rangle \neq \langle \psi | HPH | \psi \rangle$. For the qubit form of the number-projection operators we follow Ref. [36]. We see that this form can be useful in cases where one is not able to enforce the constraint from the quantum circuit (for a fault-tolerant example, in phase estimation).

TABLE I. Register-to-index map for the circuit representation of Eq. (5).

Register	Qubits	Role in Eq. (5)
Target (t)	t	Encodes occupations of the target spin orbitals; in the open-channel sector it stores the retained N -electron target state Φ_i .
Continuum (c)	c	Encodes occupations of continuum spin orbitals and the orbital index j coupled to target channel i .
Selector (a)	a	Labels computational branches by $i \in \{0, \dots, a-1\}$. In the one-hot encoding, $ i\rangle$ is the single-excitation state with only the i th selector qubit equal to $ 1\rangle$; for open channels it labels Φ_i , while for bound or L^2 branches it is completed by the occupations on the system register to identify χ_α .

B. Quantum circuits for the $N + 1$ electron problem

In standard R -matrix calculations the inner-region step amounts to solving the eigenproblem of $H_{\text{in}}(a)$ in a basis consisting of (1) antisymmetrized channel functions $\mathcal{A}\{\Phi_i u_{ij}\}$ and (2) additional L^2 configurations $\{\chi_\alpha\}$, yielding eigenpairs $\{E_k, \Psi_k\}$ and the expansion coefficients $a_{ijk}, b_{\alpha k}$ in Eq. (5). Our quantum formulation represents this same truncated basis on three qubit registers with total Hilbert space factorization $\mathcal{H} = \mathcal{H}_{\text{sys}} \otimes \mathcal{H}_{\text{sel}}$, where $\text{sys} := t+c$ and $\text{sel} := a$. We first obtain the retained N -electron target states $\{\Phi_i\}$ from a separate quantum target-state calculation on the target register. We then build the $(N+1)$ -electron circuit with $n = t + c + a$ qubits and a fixed state-preparation block V that maps computational branches to the truncated channel/ L^2 basis. In the open-channel sector of Eq. (5), the selector basis state $|i\rangle$ identifies the retained target state Φ_i and the continuum register encodes the coupled orbital index j (or more generally, a superposition over $j \in \mathcal{J}_i$), so the amplitudes of these branches are the coefficients a_{ijk} ; for bound/ L^2 branches, the selector label together with the occupations stored on the system register identifies the configuration χ_α and its amplitude $b_{\alpha k}$. The variational block $U(\theta) \in \text{SO}(k)$ acts only within this k -dimensional trial space, and at convergence its amplitudes determine the coefficients entering Eq. (5); the boundary amplitudes then follow from $w_{ik}(a) = \sum_j a_{ijk} u_{ij}(a)$, with $u_{ij}(a)$ supplied classically.

The register definitions and index conventions are summarized in Table I.

Our formalism also allows a binary selector encoding, in which the selector uses $\lceil \log_2 a \rceil$ qubits instead of a ; this reduces the qubit count but typically increases multicontrolled gate overhead. We therefore report resources below for the one-hot encoding used in our numerical experiments. Our formalism naturally allows for the inclusion of virtual orbitals (in the R -matrix sense of extra orbitals to complement the continuum description), with a modified form of the projection operator in Eq. (8). A schematic of the circuit is shown in Fig. 2.

In the close-coupling approximation used in our H_2 demonstration, each target state i couples to a single continuum orbital which we denote by a map $j(i) \in \{0, \dots, c-1\}$; more generally, each i may couple to a set $\mathcal{J}_i$ of continuum orbitals, and the circuit replaces a single controlled not (CNOT) by a controlled rotation that prepares a superposition over $j \in \mathcal{J}_i$.

The circuit decomposes as follows:

(1) A parametric mixing gate $U(\theta)$ that acts only within the selected k -dimensional trial-state manifold (implemented as a Givens-rotation cascade, Fig. 3), followed by the Clebsch-Gordan rotations $R(\zeta)$ that are fixed by spin symmetry (in our example, as in Fig. 2, a single rotation corresponding to the triplet state). The angle ζ is given by Eq. (16) and the gate is the two-qubit Givens rotation of Eq. (13) [the spin-symmetry Givens rotation derived from Eqs. (15) and (16)]. This fixed Clebsch-Gordan (CG) block prepares the spin-adapted linear combinations of target state and incoming-electron spin that define the channel functions $\mathcal{A}\{\Phi_i u_{ij}\}$ entering Eq. (5); the subsequent variational block $U(\theta)$ then mixes those channels within the trial subspace.

(2) The target-continuum couplings (blue block in Fig. 2, CNOT gates with control on the target-state selector register and target on the continuum register). Each selector qubit acts as a control on one continuum orbital qubit to ensure that, if a particular target eigenfunction is selected, exactly one continuum orbital is excited in the correct symmetry channel. In the close-coupling approximation used in our example, each target eigenfunction couples to a single continuum orbital; more generally, one can replace the CNOT by parametrized controlled rotation to mix multiple continuum orbitals.

(3) The bound-states gate (green block in Fig. 2, CNOT gates with control and targets on the target-orbital register)—a gate that entangles the selector qubits with the target-orbital qubits, so that if the circuit has selected a particular bound configuration, the correct pattern of target-orbital occupations is realized. In simple examples, one can use CNOTs for each occupied spin orbital; for larger active spaces, this becomes a cascade of controlled Givens rotations to generate the active-space correlations.

(4) The sequence of controlled selector gates, each controlled on one of the selector qubits and with targets on the target-orbital qubits.

A central element of our quantum circuits is the one-parameter two-qubit rotation operation which mixes the $|01\rangle$ and the $|10\rangle$ states, implemented through a sequence of CNOT gates interleaved with single-qubit rotations. Its matrix form is given by

$$R(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & \sin \theta & 0 \\ 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (13)$$

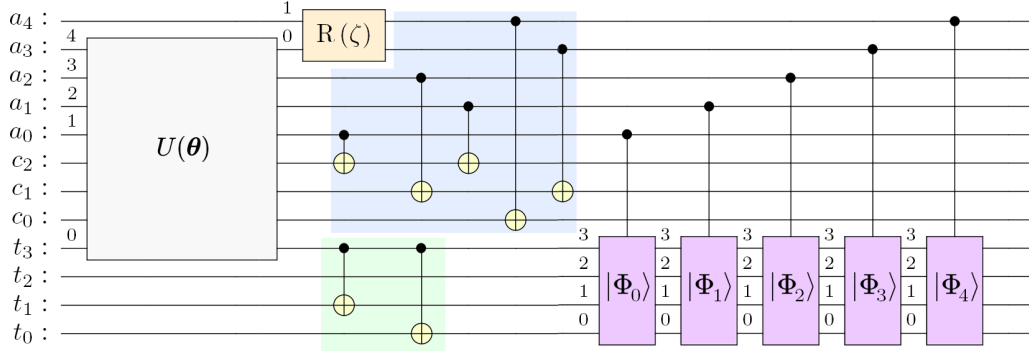


FIG. 2. Schematic of the simultaneous eigenstate circuit showing the target orbital (t), continuum (c), and target-state selector (a) registers for the symmetry subspace with $S = \frac{1}{2}$, $M = -\frac{1}{2}$, and the irreducible representation B_{1u} of the point-group D_{2h} used in the calculations. Some of the qubits that correspond to continuum spin orbitals do not appear in the circuit due to symmetry.

and it serves as the basic building block for establishing couplings in the circuit [49,51]. In practical R -matrix calculations one enlarges the trial space by increasing the number of target states, continuum spin orbitals, and L^2 configurations included in the truncation of Eq. (5). Convergence is reached by enlarging this space until the inner-region eigenvalues and boundary amplitudes entering the R -matrix sum are stable over the energy window of interest.

For the minimal H_2 demonstration (two target spin orbitals, two continuum spin orbitals and three selector qubits), the unoptimized CNOT count is 217 and the circuit depth is 314. In total, seven qubits are used, with ten variational parameters in the $SO(5)$ block, and 92 Pauli terms are measured under the subspace protocol (see Table II).

For fixed orbital bases, the total qubit count scales as $n = t + c + a$ in the one-hot selector encoding (or $n = t + c + \lceil \log_2 a \rceil$ in a binary selector encoding). The variational mixing block introduced above is implemented as a fixed cascade of $\frac{k(k-1)}{2}$ two-qubit Givens rotations (Fig. 3), so the number of entangling blocks and the circuit depth of this block scale as $\mathcal{O}(k^2)$. The target-continuum coupling layer scales as $\mathcal{O}(a)$ in the restricted close-coupling case (one continuum orbital per target state, i.e., one controlled operation per i), and more generally, as $\mathcal{O}(\sum_i |\mathcal{J}_i|)$ when each channel couples to a set $\mathcal{J}_i$ of continuum orbitals via controlled rotations. The number of distinct Pauli expectation values required to estimate $\langle H \rangle$ is set by the number of Pauli terms in the qubit Hamiltonian

after symmetry reduction; crucially, our SSO protocol avoids squared-Hamiltonian observables and requires only $\langle H \rangle$ measurements in each sequential optimization round.

Once the optimized parameters θ^* have been found, the open-channel coefficients a_{ijk} in Eq. (5) are obtained directly from the implemented variational unitary $U(\theta^*)$ restricted to the k -dimensional trial-state (channel) subspace. Concretely, we work in the single-excitation (one-hot) basis $\{|i\rangle\}$ on the selector register, so $|i\rangle$ labels the retained target channel index i . In the restricted close-coupling case used in our H_2 demonstration, each channel i is associated with a single continuum orbital $j(i)$, and the coefficient multiplying $\Phi_i u_{ij(i)}$ in Ψ_k is the amplitude of the corresponding basis state after applying $U(\theta^*)$. Equivalently,

$$a_{ij(i)k} = U_{ik}(\theta^*), \quad (14)$$

where $U_{ik}(\theta^*) \equiv \langle i | U(\theta^*) | k \rangle$ is the (i, k) matrix element of the implemented $SO(k)$ rotation in the trial-state basis. This identification is specific to the restricted close-coupling case with one continuum orbital per channel. No additional variational optimization is required to obtain a_{ijk} : once θ^* is known, the entries $U_{ik}(\theta^*)$ are fixed by the same Givens network used in the circuit and can be evaluated classically. Optionally, $|a_{ij(i)k}|^2$ can be obtained directly as the probability of the outcome [selector = i , continuum = $j(i)$] when measuring $|\Psi_k\rangle$, and relative signs or phases can be determined by a two-outcome interference measurement within a $\{|i\rangle, |i'\rangle\}$ subspace. More generally, if channel i couples to multiple continuum orbitals $j \in \mathcal{J}_i$, the circuit replaces the single controlled excitation by a controlled rotation that prepares a superposition over j , and the corresponding set of coefficients $\{a_{ijk}\}_{j \in \mathcal{J}_i}$ are obtained from the amplitudes in the joint selector-continuum single-excitation manifold, rather than from a single matrix element $U_{ik}(\theta^*)$. For larger molecules and larger trial spaces, workflow (iii) is still classical: one evaluates the same fixed sequence of Givens rotations defining $U(\theta^*)$, extracts the required amplitudes a_{ijk} in the trial subspace, and then forms $w_{ik}(a) = \sum_j a_{ijk} u_{ij}(a)$. If one reconstructs the full trial-space rotation explicitly, this post-processing scales as $\mathcal{O}(k^3)$, with an additional dependence on $\sum_i |\mathcal{J}_i|$ in the general multicontinuum case, rather than requiring an additional quantum optimization.

TABLE II. Circuit resource estimates for the minimal H_2 example. We use $n = t + c + a = 7$ qubits in the one-hot selector encoding: $t = 2$ target spin-orbital qubits, $c = 2$ continuum spin-orbital qubits, and $a = 3$ selector qubits in the circuit implementation. Together with the occupations on the system register, these selector branches generate the five trial states listed in Sec. IV. In a binary selector encoding the selector would require $\lceil \log_2 a \rceil$ qubits instead, at the expense of additional multicontrolled operations.

Number of qubits	7
CNOT gates (unoptimized)	217
Circuit depth	314
Variational parameters	10
Pauli terms measured (subspace method)	92

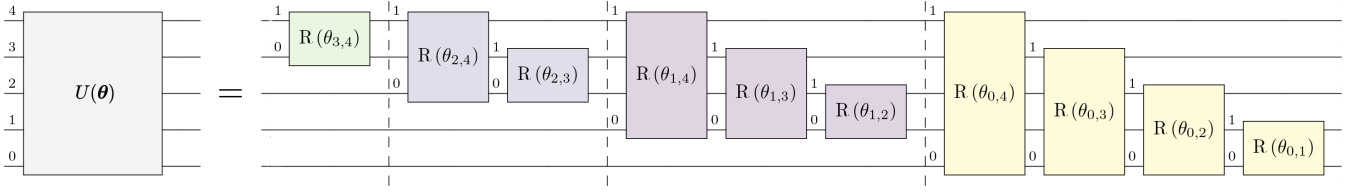


FIG. 3. Schematic of the $SO(5)$ rotation-gate decomposition into a sequence of $\frac{5 \times 4}{2} = 10$ two-qubit Givens rotations with subspace optimization.

In the simultaneous optimization algorithm, this mixing gate uses $\frac{k(k-1)}{2}$ angles (and only $k-1$ in the single-eigenstate case). The trial states include both bound configurations ($N+1$ electrons in target orbitals) and open-channel functions (N electrons in target orbitals plus one in the continuum). In our construction, only states with a single selector-qubit excitation (and similarly, only one continuum excitation) are considered, which simplifies the gate structure.

C. Spin symmetry

In the *R*-matrix formalism, we couple the target of spin S_t to an incoming electron of spin $s = \frac{1}{2}$ to a total spin S with projection M . There are four combinations for the target-state spin quantum numbers that can in principle contribute, namely, the combinations with $S_t = S \pm \frac{1}{2}$ and $M_t = M \pm \frac{1}{2}$. Whilst the coupling between the upper multiplet with $S_t = S - \frac{1}{2}$ and the lower multiplet with $S_t = S + \frac{1}{2}$ is dynamical, the couplings between states with $M_t = M - \frac{1}{2}$ and $M_t = M + \frac{1}{2}$ for total spin in the upper multiplet and the lower multiplet are each fixed by symmetry. Each spin multiplet admits a two-term expansion in terms of the half-angle ζ :

$$|S, M\rangle = \cos \zeta |S_t, M - \tfrac{1}{2}\rangle \otimes |\tfrac{1}{2}, +\tfrac{1}{2}\rangle + \sin \zeta |S_t, M + \tfrac{1}{2}\rangle \otimes |\tfrac{1}{2}, -\tfrac{1}{2}\rangle, \quad (15)$$

where ζ is determined by the Clebsch-Gordan relations for the coupling between the target eigenfunctions and the incoming electron as

$$\zeta = \begin{cases} \frac{1}{2} \arccos\left(\frac{M}{S}\right), & \text{for } S_t = S - \frac{1}{2}, \\ \frac{\pi}{2} + \frac{1}{2} \arccos\left(\frac{M}{S+1}\right), & \text{for } S_t = S + \frac{1}{2}. \end{cases} \quad (16)$$

Substituting Eq. (15) into the open-channel sector of Eq. (5), each spin-adapted channel function with total quantum numbers (S, M) takes the form

$$\begin{aligned} \mathcal{A}\{\Phi_i u_{ij}\}_{S,M} \\ = \mathcal{A}\left\{\left[\cos \zeta_i \Phi_i^{(M_t=M-\frac{1}{2})} u_{ij}^\uparrow + \sin \zeta_i \Phi_i^{(M_t=M+\frac{1}{2})} u_{ij}^\downarrow\right]\right\}, \end{aligned} \quad (17)$$

where the superscripts on Φ_i and u_{ij} indicate the target spin projection M_t and the continuum-electron spin projection m_s , respectively, and ζ_i is fixed by the target spin S_t via Eq. (16). In the quantum circuit this spin coupling is implemented by the fixed Givens rotation $R(\zeta_i)$ of Eq. (13), acting on the pair of qubits that encode M_t and m_s for channel i .

Using fixed Clebsch-Gordan rotations to enforce total-spin symmetry has known quantum-circuit realizations [52]. Our contribution is to apply a CG-enforcing circuit within a variational scattering *Ansatz*.

D. Cost functions

A critical element of our variational formulation is the choice of cost function and each optimization strategy. We consider several alternatives (where the cost functions are defined on the system register, see below for the treatment of the target-state selector register).

After Jordan-Wigner mapping, we write the system Hamiltonian as a Pauli expansion $H = \sum_\mu h_\mu P_\mu$ with $P_\mu \in \{I, X, Y, Z\}^{\otimes(t+c)}$. Expectation values such as $\langle H \rangle$ are estimated by measuring the Pauli strings P_μ and classically combining outcomes with weights h_μ . The number-projection operator P in Eq. (8) is a polynomial in Pauli- Z operators on the target or continuum registers [36], so the products HPH and (HPH) -based moments can likewise be evaluated via Pauli decomposition of the composite observable: we expand HPH into Pauli strings on the same registers and estimate $\langle HPH \rangle$ by measuring those strings (grouped into commuting sets in practice). All numerical results reported in Sec. IV use this explicit Pauli-decomposition evaluation of $\langle H \rangle$ and $\langle HPH \rangle$ on a noiseless simulator.

We denote by $\{|\psi_\mu\rangle\}_{\mu=0}^{k-1}$ the k orthonormal $(N+1)$ -electron trial basis states used in the simultaneous protocols (listed explicitly in Sec. IV). Given a shared mixing unitary $U(\theta) \in SO(k)$ acting within the span of these trial states, we define

$$|\psi_\mu(\theta)\rangle \equiv U(\theta)|\psi_\mu\rangle. \quad (18)$$

In our implementation, the full state lives on the target-orbital register (t), continuum register (c), and the target-state selector register (a), but the Hamiltonian H acts nontrivially only on the system register $\text{sys} = t+c$, i.e., as $H \otimes I_{\text{sel}}$ with $\text{sel} = a$. Here, the index μ labels trial and output states in the k -dimensional simultaneous subspace and is distinct from the channel- or target-state label i used in Eq. (5).

(1) Single state with variance cost function. In the single eigenstate approach the cost function is defined as the variance of the Hamiltonian with respect to the trial state,

$$C(\theta) = \langle HPH \rangle_{|\psi(\theta)\rangle} - \langle H \rangle_{|\psi(\theta)\rangle}^2, \quad (19)$$

where H^2 has been replaced by HPH to enforce correct occupancy. This sometimes fails to return the full eigenspectrum.

(2) Single state with folded Hamiltonian cost function. An alternative approach is to first evaluate the energy expectation for each trial state: the approximate energy $\tilde{E}$ is first evaluated for the trial state, and then used to construct a Hamiltonian [53,54] where, as long as the objective state eigenvalue is the closest eigenvalue to the energy of the trial state $\tilde{E}$,

$$C(\theta) = \langle HPH \rangle_{|\psi(\theta)\rangle} - 2\tilde{E}\langle H \rangle_{|\psi(\theta)\rangle} + \tilde{E}^2. \quad (20)$$

Here again, the projection operator P is used to enforce the occupancy constraints.

(3) Simultaneous optimization with sum of variances. In the simultaneous optimization of all five eigenstates (see explicit list in Results section), the cost function is constructed as the sum of the variances of the individual eigenstates,

$$C(\theta) = \sum_{\mu=0}^{k-1} [\langle HPH \rangle_{|\psi_{\mu}(\theta)\rangle} - \langle H \rangle_{|\psi_{\mu}(\theta)\rangle}^2]. \quad (21)$$

This approach, while necessitating separate evaluations for each eigenstate, recovers the full spectrum of that eigensector within $10^{-7} E_h$.

(4) Simultaneous optimization with subspace optimization and simple Hamiltonian expectation value. The most computational-resource-efficient method is based solely on the Hamiltonian expectation value,

$$C_{\mu}(\theta_{\mu,\mu+1}, \dots, \theta_{\mu,k-1}) = \langle H \rangle_{|\psi_{\mu}(\theta)\rangle}. \quad (22)$$

In this approach the unitary is designed to enforce a subspace structure that naturally preserves orthogonality without extra penalty terms. The overall unitary is constructed by decomposing an $\text{SO}(5)$ rotation into a sequence of ten two-qubit Givens rotations, each parametrized by an angle. The trial states are first ordered by their energy. The gate decomposition begins by coupling the second-highest eigenstate with the highest; subsequently, the third-highest eigenstate is coupled to the two highest, and so forth. Initially, all rotation parameters are set to zero to give the identity operator. The optimization is carried out in reversed order, in that the coupling parameters that connect the lowest trial state (in our example circuit $\theta_{0,1}, \theta_{0,2}, \theta_{0,3}$, and $\theta_{0,4}$) to the other trial states are optimized in the first variational round, whilst the remaining parameters are left fixed; this effectively fixes the ground-state subspace and restricts the remaining four parameters to a parametrization of $\text{SO}(4)$. Once the lowest eigenstate is determined, the same procedure is repeated for the couplings between the second-lowest trial state and the higher states (namely, $\theta_{1,2}, \theta_{1,3}$, and $\theta_{1,4}$), each time automatically constraining the evolution to a smaller subspace that remains orthogonal to the lower eigenstates. This sequential subspace optimization bypasses the need for higher-order terms such as H^2 and the explicit projection operator, and corresponds to a diagonalization of the Hamiltonian H , where at each variational round a k -dimensional block is block diagonalized into a single eigenvalue and a $k-1$ -dimensional block. Orthogonality is guaranteed by construction: at every step the *Ansatz* is a single shared $\text{SO}(k)$ rotation $U(\theta)$ acting on an orthonormal trial basis, so the prepared set $\{|\psi_{\mu}(\theta)\rangle\}$ is orthonormal for any parameter values. In the sequential subspace procedure, once the parameters that determine the lowest state are optimized, they are held fixed; the remaining free parameters then act only within the orthogonal complement subspace [equivalently, they parametrize the residual $\text{SO}(k-1)$ block], so subsequent rounds cannot reintroduce overlap with already-fixed lower states. A schematic of the decomposition of the $\text{SO}(5)$ rotation gate $U(\theta)$ is given in Fig. 3. SSVQE [34] and MCVQE [35] optimize a shared *Ansatz* over a target subspace; orthogonality is handled via state design or cost shaping, often requiring overlap measurements or explicit penalty terms. In

contrast, our SSO protocol fixes an $\text{SO}(k)$ Givens cascade that sequentially isolates eigenstates by block-diagonalizing H without penalties or Gram-Schmidt. This makes the objective purely $\langle H \rangle$ (without squared terms) and pairs naturally with our coherent-summation readout to cut measurement count.

For the algorithms 2–4 all eigenvalues are recovered within $10^{-7} E_h$.

E. Coherent summation protocol

We introduce a coherent summation readout on the target-state selector register which gives an alternative route to estimate the expectations of the operators of interest without uncomputing the selector qubits.

Let the joint state of the selector register (with a qubits) and the system register (target-orbital and continuum-orbital qubits) be

$$|\Psi\rangle = \sum_{\ell=0}^{2^a-1} |\ell\rangle_{\text{sel}} \otimes |\varphi_{\ell}\rangle_{\text{sys}}, \quad (23)$$

where $|\ell\rangle_{\text{sel}}$ are computational-basis states and $|\varphi_{\ell}\rangle_{\text{sys}}$ are the corresponding conditional system states. The goal is to access the coherently summed (generally unnormalized) system state

$$|\psi_u\rangle_{\text{sys}} \equiv (\langle u|_{\text{sel}} \otimes I_{\text{sys}})|\Psi\rangle = \frac{1}{2^{a/2}} \sum_{\ell=0}^{2^a-1} |\varphi_{\ell}\rangle_{\text{sys}}, \quad (24)$$

where we define the uniform superposition state on the selector in terms of the Hadamard gate H (not to be confused with the Hamiltonian H) as

$$|u\rangle_{\text{sel}} \equiv H^{\otimes a} |0 \dots 0\rangle_{\text{sel}}. \quad (25)$$

Equivalently,

$$|\psi_u\rangle_{\text{sys}} = (\langle 0 \dots 0|_{\text{sel}} H^{\otimes a} \otimes I_{\text{sys}})|\Psi\rangle. \quad (26)$$

We are interested in expectation values of H on the corresponding normalized postselected state.

To implement this projection using only computational-basis measurements, we apply a Hadamard gate to each selector qubit and then measure the selector in the computational basis. Postselecting the outcome $0 \dots 0$ after the Hadamard gates prepares the normalized version of $|\psi_u\rangle$ on the system register. In practice we:

- (1) apply $H^{\otimes a}$ on the selector register;
- (2) measure the selector register in the computational basis and postselect the outcome $0 \dots 0$; and
- (3) on the postselected shots, measure the Pauli strings required to estimate $\langle H \rangle$ on the system register.

Let p_0 be the probability of obtaining $0 \dots 0$ after the Hadamard gates and let $\langle H \rangle_0$ denote the conditional expectation value of H on the system given this outcome. Then,

$$\langle H \rangle_0 = \frac{\langle \psi_u | H | \psi_u \rangle}{\langle \psi_u | \psi_u \rangle}, \quad p_0 = \langle \psi_u | \psi_u \rangle, \quad (27)$$

so that $\langle \psi_u | H | \psi_u \rangle = p_0 \langle H \rangle_0$ can be reconstructed if desired.

This provides an alternative to coherently uncomputing (erasing) the selector register, which in our setting would require reversing the full selector-controlled target preparation and channel-dependent target-continuum couplings; such compute-uncompute cleanup typically increases the depth and

entangling-gate count of the relevant state-preparation portion by a factor of roughly 1.5–2 [55,56].

The postselection probability can be written as

$$p_0 = \langle \Psi | (H^{\otimes a} |0 \dots 0\rangle \langle 0 \dots 0| H^{\otimes a})_{\text{sel}} \otimes I_{\text{sys}} | \Psi \rangle \quad (28)$$

$$= \langle 0 \dots 0 | H^{\otimes a} \rho_{\text{sel}} H^{\otimes a} | 0 \dots 0 \rangle, \quad (29)$$

where $\rho_{\text{sel}} = \text{Tr}_{\text{sys}}(|\Psi\rangle\langle\Psi|)$ is the reduced state of the selector register prior to the Hadamard gates. In the one-hot encoding used throughout this work, the selector support is restricted to the Hamming-weight-one subspace spanned by $\{|e_i\rangle\}_{i=1}^a$, which implies the general bound

$$p_0 \leq \frac{a}{2^a}. \quad (30)$$

By contrast, for a binary selector encoding with $m = \lceil \log_2 a \rceil$ qubits (so $2^m \approx a$), amplitudes spread comparably over the encoded channels can yield $p_0 = O(1)$, leading to more favorable shot scaling for this postselected readout.

IV. RESULTS

To illustrate our approach, we consider the example of electron scattering from the hydrogen molecule. In this model problem the (bound) target orbitals are taken from a minimal basis as σ_g and σ_u (these orbitals are of symmetry a_g and b_{1u} in the D_{2h} point group used in the calculations and their double occupancy gives rise to states of A_g and B_{1u} symmetry), while the two continuum orbitals are chosen so that one of them is of a_g and the other one is of b_{1u} symmetry. Orbitals and integrals were obtained from a standard *R*-matrix calculation using a minimal (STO-3G) basis for H_2 , a Gaussian-type orbital (GTO) representation of the continuum, and an *R*-matrix radius of $r = 10 a_0$ using the UKRmol+ code [40].

We first solve the full N -electron target problem with $N = 2$ using the separate quantum target-state routine described in Sec. III.B, and thereby obtain the five target eigenfunctions allowed by symmetry:

$$|\Phi_0\rangle = c_0 |\sigma_g^2\rangle - c_1 |\sigma_u^2\rangle,$$

$$|\Phi_1\rangle = c_0 |\sigma_u^2\rangle + c_1 |\sigma_g^2\rangle,$$

$$|\Phi_2\rangle = \frac{1}{\sqrt{2}} (|\sigma_u^\uparrow \sigma_g^\downarrow\rangle - |\sigma_u^\downarrow \sigma_g^\uparrow\rangle),$$

$$|\Phi_3\rangle = \frac{1}{\sqrt{2}} (|\sigma_u^\uparrow \sigma_g^\downarrow\rangle + |\sigma_u^\downarrow \sigma_g^\uparrow\rangle),$$

$$|\Phi_4\rangle = |\sigma_u^\downarrow \sigma_g^\downarrow\rangle,$$

where $c_0 \approx 0.99361$ and $c_1 \approx 0.11287$. In our simulation, we consider the symmetry subspace with total spin quantum number $S = \frac{1}{2}$ and projected quantum number $M = -\frac{1}{2}$ that lies in the point-group irreducible representation B_{1u} . Here, the retained trial space has dimension $k = 5$: four of these trial states are open-channel branches built from the target eigenfunctions above, and one is the bound three-electron configuration. In the circuit implementation this trial basis is generated jointly by the $a = 3$ selector qubits and the occupations on the target or continuum registers, rather than by the selector register alone. Our circuit prepares trial wave functions corresponding to the close-coupling trial

states:

$$|\psi_0\rangle = |\sigma_g^2 \sigma_u^\downarrow\rangle,$$

$$|\psi_1\rangle = |\Phi_0\rangle \otimes |1b_{1u}^\downarrow\rangle,$$

$$|\psi_2\rangle = |\Phi_1\rangle \otimes |1b_{1u}^\downarrow\rangle,$$

$$|\psi_3\rangle = |\Phi_2\rangle \otimes |1a_{g\downarrow}\rangle,$$

$$|\psi_4\rangle = \sqrt{\frac{2}{3}} |\Phi_3\rangle \otimes |1a_{g\downarrow}\rangle - \sqrt{\frac{1}{3}} |\Phi_4\rangle \otimes |1a_{g\uparrow}\rangle.$$

The trial states $|\psi_1\rangle$, $|\psi_2\rangle$, $|\psi_3\rangle$, and $|\psi_4\rangle$ represent configurations with two electrons in a target orbital and one electron in the continuum, whereas $|\psi_0\rangle$ corresponds to the only state allowed by symmetry with all three electrons in bound orbitals. These states are each prepared by juxtaposing a one-qubit Pauli- X operator X_{q_i} to the circuit with all variational parameters set to zero.

In the eigensector with total spin $S = \frac{1}{2}$ and projection $M = -\frac{1}{2}$, the singlet target channel ($S_t = S - \frac{1}{2} = 0$) is entirely given by the $|S_t = 0, M_t = 0\rangle \otimes |\frac{1}{2}, -\frac{1}{2}\rangle$ component and no Clebsch-Gordan rotation is required. By contrast, for the triplet target channel ($S_t = S + \frac{1}{2} = 1$) the angle ζ defined by the Clebsch-Gordan relations (16) fixes the mixing between the $|S_t = 1, M_t = 0\rangle \otimes |\frac{1}{2}, -\frac{1}{2}\rangle$ and the $|S_t = 1, M_t = -1\rangle \otimes |\frac{1}{2}, +\frac{1}{2}\rangle$ components.

We perform the classical optimization using the derivative-free constrained optimization by linear approximations (COBYLA) algorithm [57,58] on a noiseless simulation. We consider a number of cost functions:

(1) Single state with variance cost function. In some instances this approach is unable to recover all the eigenvalues; for this reason we do not use it for the main benchmarks reported in Fig. 4 and focus on the three approaches below.

(2) Single state with folded Hamiltonian cost function. This approach required approximately 1397 cost-function evaluations to recover the five eigenstates, each corresponding to the evaluation of 1080 Pauli terms. The results are shown in the upper panel of Fig. 4.

(3) Simultaneous optimization with sum of variances. This method required 13 345 cost function evaluations for 1080 Pauli terms. The results are shown in the middle panel of Fig. 4.

(4) Simultaneous optimization with subspace optimization and simple Hamiltonian expectation value. This method required only 573 cost function evaluations involving 92 Pauli terms. The results are shown in the lower panel of Fig. 4.

The distinct behaviors in Fig. 4 reflect the optimization protocol: in the folded-Hamiltonian case (top) each eigenstate is optimized independently (five separate runs whose convergence traces are superimposed), whereas in the sum-of-variances (middle) and sequential subspace optimization (bottom) the parameters are optimized in a single simultaneous procedure for all five states.

The sharp dips in ΔE_k are consistent with the optimizer moving between attraction basins (and, in the SSO case, with stage-wise release or locking of different parameter subsets), leading to abrupt improvements in the targeted state's energy estimate.

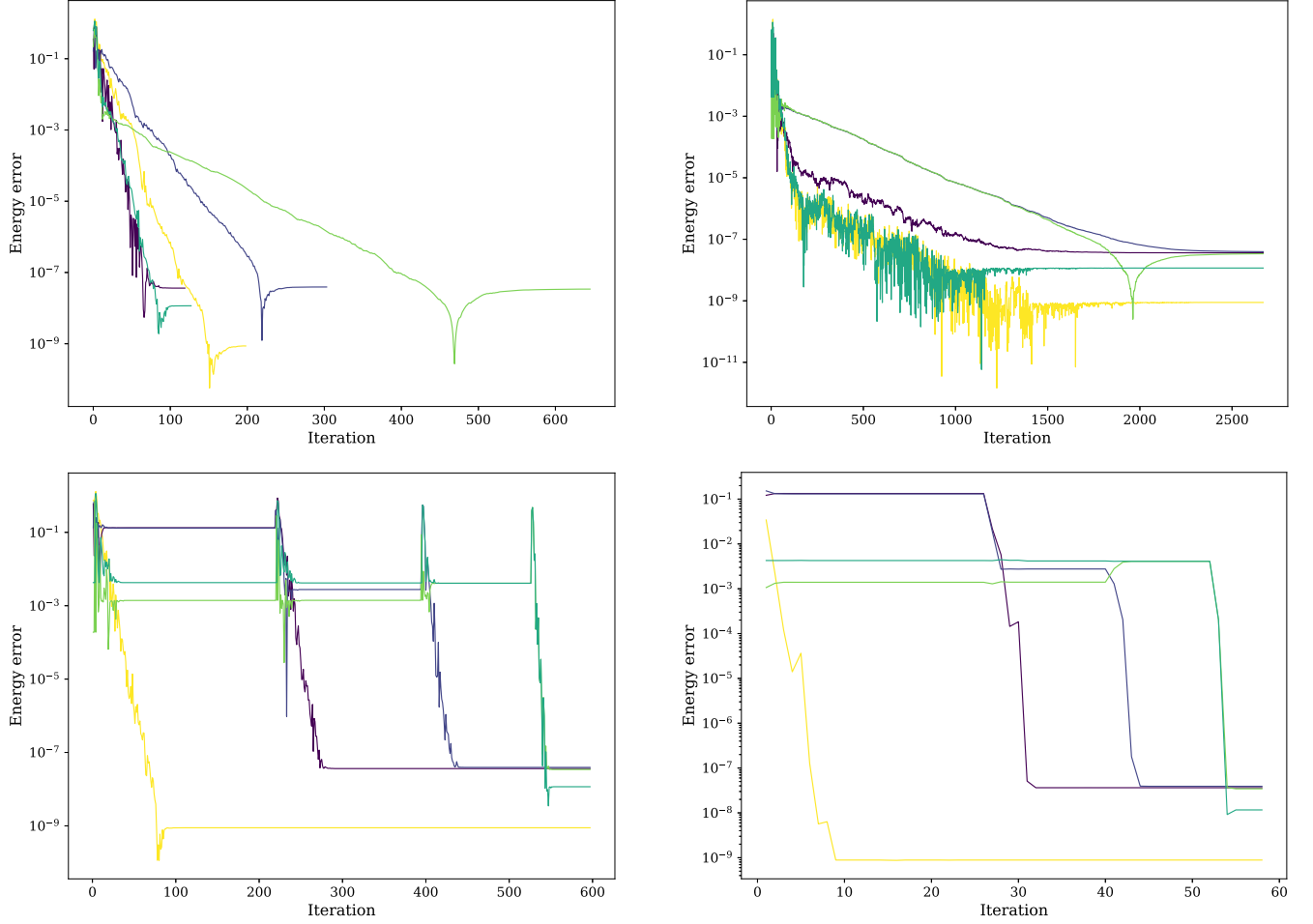


FIG. 4. Comparison of convergence behaviors for different cost functions. Here, the energy error for state k at a given optimization step is $\Delta E_k \equiv |E_k^{(\text{var})} - E_k^{(\text{ref})}|$ (in hartree), where $E_k^{(\text{ref})}$ is the corresponding exact eigenvalue obtained by classical diagonalization of the same $(N+1)$ -electron qubit Hamiltonian in the chosen symmetry sector. The $N+1$ eigenstates are labeled (in order of increasing energy) as: $|1, \frac{1}{2}, -\frac{1}{2}, B_{1u}\rangle$ (yellow), $|2, \frac{1}{2}, -\frac{1}{2}, B_{1u}\rangle$ (purple), $|3, \frac{1}{2}, -\frac{1}{2}, B_{1u}\rangle$ (dark blue), $|4, \frac{1}{2}, -\frac{1}{2}, B_{1u}\rangle$ (green), $|5, \frac{1}{2}, -\frac{1}{2}, B_{1u}\rangle$ (teal). Top left panel: Convergence diagram for single-state optimization using the folded Hamiltonian cost function. Top right panel: Convergence behavior for the simultaneous eigenstate optimization when minimizing the sum of variances. Bottom left panel: Convergence diagram for simultaneous optimization with subspace optimization and simple Hamiltonian expectation value using COBYLA. Bottom right panel: Convergence diagram for sequential subspace optimization using SLSQP with the same initial angles and stopping criterion as the COBYLA run. All curves correspond to a single optimization instance (no averaging over multiple random initializations). All variational angles are initialized to zero (so that the initial mixing block is the identity), and energies are evaluated without finite-shot sampling; the nonmonotonic features therefore reflect optimizer updates and landscape structure rather than measurement noise.

For the last three algorithms all the eigenvalues were recovered as the converged values within $10^{-7} E_h$, the residual discrepancy expected from the stopping criterion.

To assess sensitivity to the choice of classical optimizer, we repeated the SSO experiment using sequential least-squares quadratic programming (SLSQP) instead of COBYLA (with the same initial angles) and found convergence in 41 cost-function evaluations, with final energy errors of order $10^{-8} E_h$. The corresponding SSO convergence trace is shown in the bottom-right panel of Fig. 4.

V. DISCUSSION

The algorithm we have introduced fulfils several critical objectives. First, because our *Ansatz* enforces the

single-continuum occupancy constraint by construction, we can evaluate energies directly as $\langle \psi | H | \psi \rangle$ without explicitly measuring a projected observable such as PHP (or the HPH moments that arise in variance or folded-Hamiltonian costs). In our minimal H_2 instance, the plain qubit Hamiltonian requires measuring 92 Pauli strings (Table II), whereas the explicitly projected observable used in the moment-based costs expands to 1080 Pauli strings (Sec. IV), i.e., a reduction of $(1080 - 92)/1080 \approx 91.5\%$ (about $12\times$ fewer terms). The precise reduction is system dependent: it is set by how many Pauli-Z monomials appear in the projector for the chosen active space or constraint and by algebraic cancellations in the Pauli expansion of the composite observable. Second, the circuit depth is on the order of a few hundred layers for our minimal H_2 instance (depth 314 with 217 CNOTS before any

hardware-aware compilation, Table II). While this two-qubit-gate count will introduce non-negligible error on present-day NISQ hardware, the depth remains moderate compared with approaches that require deep Trotterized time evolution or phase estimation, and it can be reduced further through compilation and hardware-efficient Givens implementations. Third, the optimal rotation angles in the Givens network correspond directly to the boundary amplitudes of the *R*-matrix, thereby providing immediate access to the quantities of interest in scattering theory [39,50].

In established *R*-matrix implementations, the dominant inner-region cost is the construction and diagonalization (or iterative solution) of a large Hamiltonian in the truncated symmetry-adapted channel or L^2 basis of dimension k , which grows rapidly with the number of target orbitals, continuum orbitals, and L^2 configurations; direct diagonalization scales as $\mathcal{O}(k^3)$ time and $\mathcal{O}(k^2)$ memory, while symmetry reduction and iterative eigensolvers mitigate (but do not remove) this bottleneck. Qualitatively, as the physical description is made more realistic (more target orbitals, more continuum orbitals, more channels, and more L^2 configurations), this truncated inner-region basis size k typically grows combinatorially because it counts many-electron configurations in the chosen symmetry sector. Our approach targets this bottleneck by replacing explicit matrix diagonalization with state preparation on n qubits and expectation-value estimation of a qubit Hamiltonian, while exploiting circuit structure to reduce measurement overhead: the SSO protocol enforces orthogonality by construction and avoids H^2 -type observables, so that only $\langle H \rangle$ measurements are required per sequential round. While we do not claim a general, proven asymptotic quantum advantage over the best classical iterative methods at present, the method provides a concrete route by which quantum hardware could become advantageous as the trial-space dimension increases: the quantum device represents trial states without storing the full $k \times k$ matrix, and multiple eigenstates required for scattering can be obtained via a fixed SO(k) cascade whose parameter count and depth scale as $\mathcal{O}(k^2)$.

Among the four optimization strategies examined, the simultaneous subspace approach stands out for its efficiency. By decomposing the SO(N) rotation into a cascade of two-qubit Givens operations and enforcing orthogonality through sequential subspace optimization, we reduce both the variational parameter count and the measurement overhead. In our hydrogen-molecule test case this method recovered all eigenvalues to $10^{-7} E_h$ using only 573 cost-function evaluations and 92 Pauli terms, compared with over 10 000 evaluations for the sum of variance scheme.

Although our circuit still spans all $(N + 1)$ -electron configurations (including the bound-state sector), its structure is such that the variational parameters determine the open-channel coefficients a_{ijk} directly, so one never needs to isolate or compute the bound-state amplitudes b_{ak} . Concretely, we couple each N -electron target eigenstate only to its corresponding continuum orbital(s), and arrange the SO(k) rotation so that the resulting matrix elements yield the desired a_{ijk} . As a result, once the circuit is optimized, the coefficients a_{ijk} are obtained from the same implemented Givens network without any further variational postprocessing, even though

the full set of bound configurations appears in the Hilbert space. In this way, we avoid both deep Trotter decompositions (as in time-dependent methods) and large ancilla registers (as in quantum-walk schemes), while keeping the depth on the order of a few hundred (depth 314 for H_2 in a minimal basis, Table II) and using $\mathcal{O}(t + c + a)$ qubits in the one-hot selector encoding [or $\mathcal{O}(t + c + \lceil \log_2 a \rceil)$ with a binary selector].

Looking ahead, this framework opens several avenues for advancement. One must extend the number of continuum orbitals to obtain realistic results which may require the incorporation of more sophisticated encodings to further compress the qubit resource requirements [50]. Error mitigation techniques and hardware-efficient realizations of Givens rotations will be essential to translate these algorithms to larger systems in the form of both larger targets and more sophisticated representations of the target. The direct correspondence between circuit parameters and *R*-matrix boundary data opens the door to the integration of quantum processors into established scattering codes, offering a promising hybrid quantum classical paradigm for high-precision collision physics. Finally, our architecture shows how a CG symmetry-enforcing circuit can be integrated into variational scattering *Ansätze*, and it should transfer to other spin-coupled targets (atomic, nuclear, and model field-theory settings) [31,33].

We emphasize that, in contrast to prior quantum-computing studies of atom-molecule or nuclear scattering [30,31,33] and atomic-model electron dynamics [32], the present work targets genuine electron-molecule scattering within the *R*-matrix formalism.

VI. CONCLUSION

We have presented a variational quantum algorithm tailored to the inner-region problem of electron-molecule scattering within the *R*-matrix approach using a variational framework [9,10]. Our approach employs a number-projection operator to enforce physical occupancy constraints, a rotation-based *Ansatz* to generate the required symmetry-adapted subspaces, and a coherent summation protocol to simplify observable estimation. Benchmarking on the hydrogen-molecule system demonstrates that all scattering eigenvalues in a given symmetry sector are obtained while using a shallow circuit and minimal measurement resources. By making the open-channel coefficients, and hence the *R*-matrix boundary amplitudes, directly accessible from the optimized circuit, our method provides straightforward access to scattering observables. This work represents a significant step toward the practical application of quantum computers in molecular collision studies and lays the groundwork for future extensions to more complex targets and continuum representations.

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D.P. conceived and developed the central ideas, including the Hamiltonian formulation, circuit, and algorithm design. V.G. provided the molecular integrals from the UK Molecular R-matrix Codes (UKRmol+) used to construct the Hamiltonian and contributed significantly to the discussion. J.T. and J.D.G. offered valuable insights and discussions. All authors contributed to the manuscript.

DATA AVAILABILITY

The code that implements the variational algorithms described in this work, together with the molecular integrals and per-iteration energies needed to reproduce the four numerical experiments reported in Sec. IV, is openly available in a zenodo-archived GitHub repository [59].

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